

Chapter 70 — Surgical Management of Male Infertility

- **American Urological Association (AUA) / American Society for Reproductive Medicine (ASRM) joint guidelines (2021)** set evidence-based standards for the male partner of the infertile couple: diagnostic testicular biopsy, varicocelectomy, reconstruction for obstructive azoospermia (OA) and indication-based sperm retrieval.
- **In vitro fertilisation (IVF) with intracytoplasmic sperm injection (ICSI)** treats even the most severe male factor — unreconstructable obstruction and nonobstructive azoospermia (NOA) — but is costly and burdens the female partner: ovarian hyperstimulation, multiple gestation, complications of oocyte retrieval.
- **ICSI bypasses all natural biological barriers:** realistic concern of transmitting genetic abnormalities, and an association with increased birth defects.
- **Specific male treatment** — microsurgical reconstruction for OA, varicocelectomy for impaired testicular function — in properly selected patients remains the **safest and most cost-effective** management.
- Treating the man can **upgrade the couple's pathway:** donor sperm → IVF/ICSI with testicular microdissection → IVF/ICSI with ejaculated sperm or intrauterine insemination (IUI) → natural conception.
- **Microsurgical testicular sperm extraction (microTESE)** makes retrieval, fertilisation, pregnancy and live birth feasible in NOA; retrieval now spans testis, epididymis and seminal vesicle, percutaneous or open.
- The **operating microscope identifies individual seminiferous tubules likely to contain sperm** — higher TESE success with less morbidity; microfluidics is under investigation to improve yield and efficiency.
- **Varicocele causes progressive, duration-dependent testicular injury.** Microsurgical varicocelectomy, once reserved for oligospermia, has induced spermatogenesis and returned sperm to the ejaculate in some men with NOA.
- Varicocele also impairs **Leydig cell function** (lower serum testosterone than age-matched controls); repair halts and partly reverses the decline → emerging concept of **early repair to prevent future infertility and androgen deficiency.**

Concept — *Safer microsurgery has moved the urologist **from salvaging residual testicular function to preventing its loss** — and correcting the male factor can move a couple down the ladder of assisted reproduction.*

SURGICAL ANATOMY

- The scrotum is uniquely accessible to examination, imaging and surgery; **scrotal examination is a lost art** — no substitute for history and physical examination.
- Success rests on **selecting the right operation for the correct indication**, which requires anatomy (Chapter 66) and physiology (Chapter 67).

Testicular Blood Supply

- Three sources: **testicular (internal spermatic) artery from the aorta — the main supply; deferential (vasal) artery** from the internal iliac (hypogastric) artery or its superior vesical branch; **cremasteric (external spermatic) artery** from the inferior epigastric artery, chiefly to the tunica vaginalis with some testicular branches.
- **Testicular artery diameter exceeds deferential plus cremasteric combined.**

- Iatrogenic risk: **inguinal hernia repair, orchidopexy, vasectomy**. After inadvertent testicular artery ligation collaterals may perfuse, yet **atrophy, chronic pain and/or azoospermia** may follow in adults and children.
- **After vasectomy the vasal artery is presumed compromised** — the testicular artery must be preserved in any later operation, e.g. varicocelectomy.

Epididymal Blood Supply

- **Superior and medial epididymal arteries from the testicular artery**; cauda (inferior pole) from the **vasal (deferential) artery**.
- The superior and inferior supplies **interconnect extensively**: after vasectomy the testicular contribution is more than adequate.
- For vasoepididymostomy or vasovasostomy the epididymis can be dissected off the testis up to the caput with the inferior and medial epididymal arteries ligated — **safe provided the superior epididymal artery is intact**; this yields an additional 4–6 cm of length.

Blood Supply of the Vas Deferens

- **Abdominal (seminal vesicle) end: deferential artery. Testicular end: deferential artery plus inferior epididymal interconnections**; the two freely anastomose.
- After vasectomy the testicular end is supplied **entirely from testicular/epididymal branches**; the abdominal end **entirely from the deferential artery**.
- The vas receives **no supply from cremaster or other cord vessels** → if transected or obstructed at two sites with vasal vessels interrupted at both, the **intervening segment fibroses**.
- Hence **never two simultaneous vasovasostomies on one vas**; excise the intervening segment and perform a single tension-free anastomosis.

TABLE 70.1 BLOOD SUPPLY TO TESTIS, EPIDIDYMIS, AND VAS DEFERENS

Structure	Artery	Origin / territory
Testis	Testicular (internal spermatic) — main	Aorta
	Deferential	Internal iliac / superior vesical
	Cremasteric (external spermatic)	Inferior epigastric; mainly tunica vaginalis
Epididymis	Superior (and medial) epididymal	Testicular artery — caput
	Inferior epididymal	Vasal (deferential) artery — cauda
Vas deferens	Seminal vesicle (abdominal) end	Deferential artery only
	Testicular end	Deferential + inferior epididymal

- **What the table tests**: every structure has dual supply except the post-vasectomy vas segments; the trap is assuming cord or cremasteric vessels feed the vas — they do not.

Anatomy of the Excurrent Ducts

Rete testis (intratesticular, beneath tunica albuginea at the cranial pole, near arterial entry) → **7–11 efferent ducts** → convolute and freely anastomose to form the **caput** → coalesce at distal caput into a **single tubule from caput–corpus junction** → vas → urethra

- **Injury or ligation distal to the caput obstructs the entire system on that side**; a **single stitch through a corpus or cauda tubule** causes complete obstruction.
- Iatrogenic epididymal injury: **hydrocelectomy and spermatocelectomy** (long-standing large/tense hydrocele splays the epididymis), **post-infection** operations, **orchidopexy for torsion**.

- Protection: operating microscope and **transillumination of the hydrocele sac; generous margins from the epididymis** at hydrocelectomy.
- **Avoid spermatocelectomy in men wanting fertility** unless it is the suspected cause; then use a microsurgical approach.
- Sperm in the caput are **functionally immature** (late maturation occurs there) → **pregnancy lower when vas is joined to caput** than to corpus or cauda.

CAPUT VERSUS CORPUS/CAUDA — PUNCTURE AND INJURY

	Caput	Corpus / cauda
Tubule architecture	Multiple lobules, anastomosing efferent ducts	Single tubule
Single puncture for aspiration	Safe at most proximal caput	Obstructs
Multiple punctures	Obstruct	Obstruct
Sperm maturity	Immature	Mature; better pregnancy after anastomosis

Ejaculatory Ducts

- Left and right ducts enter the **prostatic urethra at the level of the utricle; bilateral ejaculatory duct obstruction (EDO) → azoospermia**.
- Causes: **congenital** (midline cyst), **acquired** (genitourinary infection), **iatrogenic** (urethral instrumentation).
- Semen: **azoospermia or severe oligospermia, low volume, acidic pH, negative/equivocal/low fructose**.
- **Transrectal ultrasound (TRUS)** shows dilated seminal vesicles and/or ejaculatory ducts; **sperm on seminal vesicle aspiration in an azoospermic man confirms EDO/atresia**.
- **Transurethral resection of the ejaculatory ducts (TURED)** relieves obstruction but is **not benign**: it destroys the valve-like mechanism that prevents urine reflux → chemical/bacterial epididymitis. **Seminal vesiculoscopy** is gaining traction to limit this.

KEY POINTS

- The testicular artery is the main blood supply to the testis.
- If the epididymis is injured or ligated distal to the caput, the entire system on that side is obstructed.
- Suspect bilateral EDO in an azoospermic man with low-volume ejaculate and dilated seminal vesicles on imaging.

Concept — Two anatomical facts generate most of the surgical rules: a **single dominant testicular artery** that must be preserved once the vasal artery is lost, and a **single epididymal tubule distal to the caput** in which one stitch or puncture is enough to obstruct.

TESTIS BIOPSY

Indications

- Indicated in the **azoospermic man with normal testicular size and consistency, palpable vasa and normal follicle-stimulating hormone (FSH)** — distinguishes OA from NOA (AUA/ASRM guideline).
- **Cryopreserve at biopsy** → avoids a repeat retrieval at the time of IVF.

- **Biopsy unnecessary: congenital bilateral absence of the vas deferens (CBAVD) with normal FSH** (spermatogenesis always present) before definitive retrieval; **highly positive serum antisperm antibody** with a history of iatrogenic vasal (hernia) or epididymal (hydrocelectomy/spermatocoelectomy) injury — obstruction almost certain.
- **Diagnostic biopsy is usually bilateral** irrespective of size discrepancy.
- Because ICSI needs only a single sperm, biopsy is **diagnostic and potentially therapeutic**; even men with **markedly elevated FSH and small soft testes** often harbour rare mature sperm.
- NOA testes are **heterogeneous** and testicular sperm motility can be induced → examine **fresh unfixed tissue** for tailed, possibly motile sperm, and multiple samples if initially negative; an **andrology laboratory must be available** to process and cryopreserve.
- **Biopsy alone is rarely indicated** — usually planned as TESE.
- Frozen retrieved sperm do not compromise IVF/ICSI, **except in spinal cord injury (SCI)**, where a fresh testicular sperm-ICSI cycle may be better.

Open Testis Biopsy

- **Open biopsy with TESE is the gold standard** — optimal tissue for diagnosis and sperm retrieval.
- General, spinal or local anaesthesia; **local with spermatic cord block** gives pain relief, office convenience and cost-effectiveness. Loupes or microscope show vessels on the tunica albuginea.
- **Choose the larger testis**; with unilateral varicocele or prior varicocelectomy use the **contralateral** testis.

Assistant stretches scrotal skin over anterior testis, epididymis posterior → (local) 3–4 cm wheal + cord block 5–10 mL → **2 cm transverse incision in skin folds** (or 3–4 cm median raphe for bilateral, with two separate dartos openings) → skin and dartos, haemostasis → curved haemostat / Jacobson forceps to tunica vaginalis, open → **Barraquer eyelid retractor** → enlarge and deliver testis if anatomy distorted, epididymis not palpable posteriorly or tunica albuginea unclear → tunica edges held; **wound dry before incising albuginea** → **two 5-0 Vicryl stay sutures parallel to surface vessels** → **3–4 mm incision with 15-degree microknife** → bipolar to crossing vessels → **pea-sized sample excised with iris scissors**

- **Minimal handling** (no forceps) for fixed tissue; fixatives: **Bouin's, Zenker's or collidine-buffered glutaraldehyde**. **Never formalin** — it distorts testicular histology.
- **Touch prep**: blot cut surface on slide + drop of saline, lactated Ringer's or human tubal fluid (HTF) + coverslip → high power ± phase contrast shows **tailed sperm and motility**. Negative → **wet squash prep** (specimen crushed under coverslip). Negative → close and biopsy contralateral testis.
- **Microscope at 10–25× allows selective sampling of larger tubules** more likely to contain sperm.
- Sperm found → slide and extra tissue to andrology for cryopreservation; **nonabsorbable suture marks the site of spermatogenesis** for future TESE.
- Closure: tunica vaginalis continuous 5-0 Vicryl (**limits scarring for future surgery**) → dartos running 4-0 Vicryl → skin 4-0 Monocryl horizontal mattress → bacitracin, fluffs, snug support. **No antibiotics**.
- **Tips**: loupes/microscope and Barraquer retractor; avoid transecting albuginea vessels and keep specimen blood-free; preplaced parallel sutures give upward traction and easy closure; close tunica vaginalis to prevent albuginea-tissue scarring before repeat biopsy.

Percutaneous Testis Biopsy

- **14-gauge prostate biopsy gun** — a **blind procedure** with risk to epididymis or testicular artery; **not after surgery that has scarred the anatomy**. Cord block needed under local.
- Therapeutically, **percutaneous biopsy or aspiration is most useful for fresh retrieval in OA with normal spermatogenesis**.

Percutaneous Testicular Aspiration

- **23-gauge needle or angiocatheter sheath:** less invasive and painful than percutaneous biopsy, but **few tubules with poorly preserved architecture.**
- Flow cytometry distinguishes haploid from diploid cells (late spermatogenesis), but **direct wet examination for sperm and motility is the practical test;** 3–4 aspirations until sperm found.
- In OA, aspirated sperm serve ICSI **when epididymal retrieval fails; fine-needle aspiration yields far fewer sperm than microTESE in NOA.**

OPEN BIOPSY VERSUS PERCUTANEOUS BIOPSY VERSUS ASPIRATION

	Open biopsy / TESE	Percutaneous core biopsy	Testicular aspiration
Instrument	15° microknife, iris scissors	14-G biopsy gun	23-G needle / angiocath
Vision	Direct ($\pm$ microscope)	Blind	Blind
Tissue / histology	Optimal; gold standard	Core	Few tubules, poor architecture
Best use	Diagnosis + retrieval; NOA	Fresh retrieval in OA	Fresh retrieval in OA; backup to epididymal
Avoid	—	Scarred anatomy	NOA (low yield)

Complications of Testis Biopsy

MOST SERIOUS VERSUS MOST COMMON

Most serious — inadvertent epididymal biopsy	Most common — haematoma
Epididymis with sperm in tubule on histology → obstruction at that site is certain; epididymis without sperm → obstruction above that level or primary failure — no harm done	Can be large; drainage exceedingly rare, especially with open biopsy; prevented by magnification to avoid vessels and bipolar haemostasis; continuous 5-0 closure of the vascular tunica vaginalis limits bleeding and adhesions

- Wound infection is rare without haematoma; **antibiotics unnecessary.**

KEY POINTS

- Biopsy is diagnostic and potentially therapeutic and can be performed safely under local anaesthesia in the office.
- Percutaneous biopsy or aspiration is most useful for fresh sperm retrieval for IVF/ICSI in OA with normal spermatogenesis.
- Open testis biopsy provides the most consistent and highest yield of all sperm retrieval techniques.

Concept — *Because one sperm can be the treatment, **biopsy has become a retrieval operation** — fresh examination, an on-site laboratory, cryopreservation and a marked site all follow from that.*

VASOGRAPHY

Indications

ABSOLUTE VERSUS RELATIVE INDICATIONS

Absolute — all three required	Relative
Azoospermia	Severe oligospermia with normal testis biopsy

Absolute — all three required	Relative
Complete spermatogenesis with many mature spermatids on biopsy	High sperm-bound antibodies (unilateral, bilateral or partial obstruction)
At least one palpable vas	Low semen volume + very poor motility (partial EDO)

- Vasography asks two questions: **are there sperm in the vasal fluid, and is the vas obstructed?**

VASAL FLUID FINDINGS WHEN BIOPSY SHOWS MANY SPERM

Vasal fluid	Meaning / next step
No sperm	Obstruction on testicular side — most likely epididymal ; motile sperm in epididymal fluid at vasoepididymostomy confirm. No sperm in epididymal tubules → efferent ductule / rete obstruction
Copious, many sperm	Vasal obstruction or EDO → formal contrast vasography to locate
Copious thick white, no sperm, dilated vas	Secondary epididymal obstruction in addition to vasal obstruction or EDO
Many sperm, normal vasogram	Retrograde ejaculation, failure of emission or vasal aperistalsis

- Contrast vasography with intraoperative radiographs helps after **inguinal hernia repair or pelvic surgery** to judge reconstructability.
- Ideally **same sitting as biopsy and any vasoepididymostomy** — scar makes later vasoepididymostomy harder; careless technique causes **stricture or obstruction at the vasography site**.
- **Vasography cannot diagnose epididymal obstruction** — and most non-vasectomy obstructions are epididymal.
- Normal spermatogenesis + palpable vasa → vasography, if needed, **only at the time of definitive repair**.

Technique of Vasography and Interpretation of Findings

- **Inguinal hernia repair, particularly in childhood**, is a known cause of vasal injury.

No prior inguinal incision and site unknown → **high vertical scrotal incision**, external ring marked → vas isolated at **straight-convoluted junction** → 10×: sheath incised longitudinally, **vasal vessels preserved** → bare vas on vessel loop, straight clamp as platform → 25×: **15° microknife hemitranssection** to lumen → fluid on slide + saline + coverslip; repeat after milking epididymis and convoluted vas (persistently spermless → **epididymal obstruction**) → motile sperm: **barbotage testicular end with 0.2 mL HTF and cryopreserve** — **before any indigo carmine or contrast** → abdominal end cannulated with **24-G angiocatheter sheath, 1 mL saline** via 1-mL tuberculin syringe → passes easily = patent, formal vasography unnecessary → (very rarely) 1 mL 50% indigo carmine with bladder catheter: blue/green urine confirms patency

INDIGO CARMINE VERSUS METHYLENE BLUE

Indigo carmine (diluted 50/50)	Methylene blue
Preferred for patency testing	Kills sperm even at low concentration — useless for cryopreservation or immediate ICSI

Indigo carmine (diluted 50/50)	Methylene blue
Used before sperm are secured	Only after sperm aspirated and cryopreserved (e.g. to aid TURED)

- **Copious brownish 'crank-case oil' fluid with sperm, markedly dilated vas** → obstruction toward seminal vesicle end → pass **2-0 Prolene** toward seminal vesicle; clamp where it stops — delineates inguinal obstruction from prior groin surgery.
- Obstruction proximal to inguinal scar → **formal vasography via No. 3 ureteral catheter**; optional **16-Fr Foley, balloon 5 mL air on gentle traction** prevents contrast reflux obscuring detail and marks the bladder neck → **0.5 mL water-soluble contrast**.
- Hemitranssection sites closed microsurgically: **2-3 interrupted 10-0 nylon for mucosa, 8-0 for muscularis/adventitia**.

VASOGRAM FINDINGS AND ACTION

Finding	Interpretation	Action
Obstruction at ejaculatory ducts	EDO	Indigo carmine (or methylene blue after sperm banked) into both vasa → TURED
Both vasa opacify from one injection	Common cavity	Usually midline ejaculatory duct cyst
Vas ends blindly far from ducts	Congenital partial absence of vas	Cystic fibrosis mutation testing ; bilateral → reconstruction impossible, aspirate and cryopreserve for ICSI
Inguinal obstruction	Iatrogenic	Inguinal vasovasostomy or crossed transeptal vasovasostomy
No sperm, patent seminal vesicle end	Epididymal obstruction	Complete transection → vasoepididymostomy
Many sperm, normal vasogram	Functional	Retrograde ejaculation / failure of emission / aperistalsis

Complications of Vasography

- **Stricture: repeated percutaneous sharp-needle attempts**, imprecise vasotomy closure, **non-water-soluble contrast** (never use).
- **Injury to the vasal blood supply**: vasovasostomy proximal to an injured vasography site → **ischaemia, necrosis, obstruction of the intervening segment**.
- **Haematoma**: meticulous **bipolar** haemostasis in the perivasal sheath.
- **Sperm granuloma**: leaky closure → granuloma → stricture/obstruction; closure technique identical to vasovasostomy.

Transrectal Vasography and Seminal Vesiculography

- **Markedly dilated seminal vesicles and/or midline Müllerian duct cyst** in OA → **TRUS-guided aspiration then indigo carmine with contrast**; same bowel preparation and antibiotics as TRUS prostate biopsy.
- **Sperm in aspirate** → **at least one vas and epididymis patent** → 0.5 mL indigo carmine + 1.5 mL 50% water-soluble contrast → resectable lesion → TURED; **blue efflux** from ducts or unroofed cyst confirms adequate resection.
- **Obviates open scrotal vasography** for transrectally accessible lesions; TUR can proceed without violating the scrotum; aspirated sperm frozen in case surgery fails.

- **No sperm → secondary epididymal obstruction.** Simultaneous TURED + vasoepididymostomy is **rarely successful** → EDO with bilateral epididymal obstruction is best managed by **sperm retrieval and cryopreservation for ICSI.**

Summary

- Vasography **only if biopsy confirms spermatogenesis consistent with OA.**
- **Only at the time of planned reconstruction.**
- **Vasal fluid always sampled first** to allow cryopreservation.
- **Indigo carmine, not methylene blue,** to confirm patency.
- Formal x-ray vasography only to locate obstruction **proximal to the internal inguinal ring.**
- Dilated seminal vesicles or midline cyst → **transrectal aspiration with contrast and indigo carmine;** cryopreserve motile sperm.

KEY POINTS

- Perform vasography only at the time of planned reconstruction.

Concept — Vasography interrogates the **downstream** tract, so it is justified only once sperm production upstream is proven and only when the answer can be acted on **in the same sitting**; the vasal fluid sample, taken first, often answers the question on its own.

VASOVASOSTOMY

- **About 500,000 vasectomies a year in the United States,** stable; divorce rate 40–50%; a small proportion ultimately seek reversal and a larger minority express interest in future fertility.
- **Iatrogenic vasal injury,** usually from hernia repair, causes OA in a proportion of azoospermic men.

Preoperative Evaluation

- Document **adequate spermatogenesis:** prior natural fertility before vasectomy usually suffices; **FSH** if significant medical change, bilateral small testes or **advanced paternal age.**

Physical Examination

EXAMINATION FINDINGS AND THEIR MEANING

Finding	Significance
Small or soft testes	Impaired spermatogenesis; poor outcome
Indurated, irregular epididymis	Secondary epididymal obstruction → vasoepididymostomy likely
Sperm granuloma at testicular end	Leak vented epididymal pressure → better prognosis regardless of interval
Large vasal gap / high vasectomy	May need inguinal extension ; testis may ride high → chronic pain risk
Inguinal or scrotal scars	Iatrogenic inguinal (hernia) vasal or epididymal (hydrocelectomy, orchidopexy) obstruction

Laboratory Tests

- **Centrifuged semen pellet:** complete tailed sperm are found in some men many years after vasectomy → sperm certain in the vas on at least one side — **favourable prognosis. Hypospermia → TRUS for additional EDO.**
- **Serum antisperm antibodies** corroborate obstruction and active spermatogenesis but are **not required.**

- **FSH** in small, soft testes; **elevated FSH predicts impaired spermatogenesis and poorer prognosis.**

Anaesthesia

- **General anaesthesia preferred** — small movements are magnified by the microscope. Regional or local with sedation in cooperative patients per surgeon experience.
- Operate only if **capable of vasoepididymostomy**; incisions may need extension for large gaps; vasoepididymostomy can take **about 4 hours**.
- **Hypobaric spinal with long-acting bupivacaine gives 4–5 hours** and abolishes lower-body motion; epidural catheter equally effective.
- **Local with liposomal bupivacaine + sedation** is workable for **small gaps and <10 years since vasectomy**.

Surgical Approaches

SCROTAL VERSUS INGUINAL APPROACH

Scrotal	Inguinal
Bilateral high vertical incisions or single 4 cm midline raphe incision	Preferred when inguinal vasal obstruction from herniorrhaphy or orchidopexy is strongly suspected
Length is a problem on the abdominal end, not the testicular end ; large gap/high site → Y-extension toward the external ring ; low site → pull up testicular end	Incision through old scar leads directly to obstruction; if lesion proves scrotal/epididymal, deliver testis through same or separate scrotal incision
Deliver testis only if vasoepididymostomy needed or site in convoluted vas near epididymal tail; leave tunica vaginalis intact , and close it if opened or testis may not return to anatomical position	Intra-abdominal robotic approach after laparoscopic/robotic inguinal hernia repair

Preparation of the Vasa

Vas clamp above obstruction; penetrating towel clamp secures tissue **excluding the cord** → sharp dissecting forceps develop plane under vas, **perivasal sheath and vasal vessels preserved** → vessel loop; two-finger window → transilluminate sheath to see vessels → mobilise for tension-free anastomosis, **never stripping the sheath** → excise obstructed segment and any granuloma **staying directly on vas** (protects testicular artery — injury here → **atrophy, as the vasal artery is already interrupted**) → blunt finger/gauze mobilisation of abdominal vas to external ring, with haemostasis → through external ring to internal ring if needed

LENGTH-GAINING MANOEUVRES, IN ESCALATING ORDER

Manoeuvre	Gain / note
Blunt dissection to external ring, then to internal ring	Vasal vessels left intact
Free entire convoluted vas from epididymal tunica (testis drops upside down)	4–6 cm ; loupes or microscope to protect vessels
Extend to internal ring, divide inguinal floor, reroute vas beneath it (as in difficult orchidopexy)	Shortens path
Dissect epididymis off testis from vasoepididymal junction to caput	4–6 cm ; superior epididymal vessels preserved supply testicular end
All combined	Gaps up to 10 cm bridged

- **Testicular end cut transversely**: No. 11 blade through slotted 2-, 2.5- or 3-mm nerve-holding clamp (or over Adson handle) — **one smooth 90° cut, no sawing**.

- **Healthy end:** white mucosal ring that **springs back after gentle dilation**; smooth, soft muscularis; **bullseye** of three layers; bleeding from mucosal edge and muscularis. **Gritty muscularis = scar → recut**; poor bleeding → recut. Bleeders: microbipolar at low power.
- Milk vas, touch slide, mix with saline/Ringer's, coverslip. No fluid → **aspirate** with 24-G sheath.

TESTICULAR END VERSUS ABDOMINAL END

Testicular end	Abdominal end
Aspirate only — never flush: high pressure damages epididymal tubules	Dilate with microvessel dilator, cannulate 24-G, flush 1–3 mL saline to prove patency
Fluid examined for sperm → decides vasovasostomy versus vasoepididymostomy	Awake patient feels perineal pressure / urge to void; resistance = obstruction → vasogram or 5-0 Prolene threaded to resistance, clamped to measure

- **Minimal instrumentation of the mucosa.** Ends approximated tension-free with **interrupted 5-0 Prolene through the sheath**; tongue blade covered by Penrose or flat drain as platform; fine clamps fix sheath to backboard.

When to Perform Vasoepididymostomy

TABLE 70.2 RELATIONSHIP BETWEEN GROSS APPEARANCE OF VASAL FLUID AND MICROSCOPIC FINDINGS

Vasal fluid	Microscopy	Procedure
Copious, crystal clear, watery	No sperm	Vasovasostomy
Copious, cloudy, thin, water-soluble	Sperm with tails (incl. short tails), motile or not	Vasovasostomy
Copious, creamy yellow, water-insoluble	Many sperm heads, acrosome often visible	Vasovasostomy
Scant fluid, granuloma at vasectomy site	Barbotage shows sperm, usually tailed, often motile	Vasovasostomy
Copious, thick white toothpaste-like , water-insoluble	No sperm or heads	Vasoepididymostomy
Scant white thin fluid	No sperm or heads	Vasoepididymostomy
Scant fluid, no granuloma	No sperm or heads	Vasoepididymostomy
Any fluid	Occasional grape-like clusters of sperm heads	Vasoepididymostomy

- **What the table tests:** gross appearance predicts microscopy; **sperm or heads → vasovasostomy**, with two exceptions — **crystal-clear watery fluid without sperm still gets vasovasostomy** (sperm likely to return), and **grape-like clusters of heads mean epididymal obstruction**.
- Sperm with tails → vasovasostomy. No fluid → **barbotage with 0.1 mL saline via 24-G sheath while vigorously milking the convoluted vas**.
- **Large granulomas:** little dilation and scant fluid, but barbotage **invariably finds sperm** → vasovasostomy.
- No granuloma and **dry, spermless on multiple samples** → vasoepididymostomy.
- **Toothpaste fluid** → open tunica vaginalis and inspect: clear epididymal obstruction (epididymal granuloma, dilated tubules above, collapsed below) → vasoepididymostomy.
- **When in doubt, or inexperienced with vasoepididymostomy, perform vasovasostomy** — though few men with bilaterally spermless vasal fluid after intensive search regain ejaculated sperm after it.

Multiple Vasal Obstructions

- Abdominal end not patent → thread **2-0 nylon or polypropylene** to locate second obstruction.
- **Within 5 cm of vasectomy site** → dissect abdominal vas to that site and excise; extend inguinally to free vas toward internal ring; free testicular end to vasoepididymal junction.
- **Too far for one anastomosis** and contralateral testis lacks sperm or is not easily reconstructed → **single crossed vasovasostomy** to create one good system; otherwise aspirate vasal/epididymal sperm into micropipettes and cryopreserve.
- **Never simultaneous vasovasostomies at two sites** → devascularisation, fibrosis and necrosis of the intervening segment.

Varicocelectomy and Vasovasostomy

- Proper varicocelectomy ligates all spermatic veins, leaving **vasal veins as the only venous return**; after vasectomy and reversal **vasal veins and vasal artery are likely compromised** → must preserve the **testicular artery** and **some venous return**.
- **Microscopic varicocelectomy** preserves the artery; deliberate preservation of small **cremasteric or perivasal veins** maintains drainage.
- In one series, leaving cremasteric and periarterial veins intact gave a **clearly higher recurrence** than standard microsurgical varicocelectomy in non-vasectomised men → these veins drive a significant share of recurrences. Another series combined both without preserving them: low recurrence, no atrophy.
- If combined: **microscope and testicular artery preservation mandatory**. Alternative, especially with a young female partner: **reconstruct first, assess semen, then varicocelectomy ≥6 months later** once venous and arterial channels cross the anastomosis — two-stage approach without atrophy or recurrence.

Anastomotic Techniques: Keys to Success

SIX PRINCIPLES OF TUBULAR ANASTOMOSIS

Principle	Why
Accurate mucosa-to-mucosa	Testicular lumen often 2-5 times the abdominal diameter; equal-lumen techniques fail on discrepant lumina
Leakproof	Sperm are highly antigenic; extravasation inflames; vasal/epididymal fluid lacks platelets and clotting factors , so watertightness depends entirely on mucosal sutures
Tension-free	Tension → early sperm, then falling count/motility → azoospermia; only a thin fibrotic band at re-exploration; prevent by freeing vasa and sheath reinforcing sutures
Good blood supply	Recut until healthy bleeding; gain length as described
Healthy mucosa and muscularis	Recut if mucosa poorly distensible, peels or shreds, or muscularis gritty; needle electrocautery vasectomy damages far beyond the tip
Atraumatic technique	Multiple errors (mucosal cuts, tear-through, back-walling) → resect and redo immediately

Setup

- **Operating microscope 6-32×**, diploscope (identical fields), foot-pedal zoom and focus.
- **3D 4K exoscopes** (ORBEYE, VITOM, NGENUITY) let the surgeon sit upright viewing a screen — **less fatigue** in long cases.
- **Microsurgical chairs stabilising chest and arms**; cheap alternative: rolling stool with taped bean bag, arm boards built up with blankets.

- **Right-handed surgeon sits on the patient's right** so the forehead stitch falls on the **smaller, harder** abdominal lumen.

Microsurgical Multilayer Microdot Method

- Handles **markedly discrepant lumina** in straight or convoluted vas; **separates planning from placement** by mapping each suture.
- **Exactly six mucosal sutures** for every anastomosis — easy to map and always leakproof. Microtip marking pen (Devon Skin Marker Extra Fine #151).

Dry testicular end (Weck-cel) → dot at **3 o'clock halfway between mucosal ring and outer muscle edge**, reference line → 9 o'clock + line → 11, 1, 5, 7 (six) → abdominal end marked to match → **double-armed 10-0 monofilament nylon, 70-μm taper needles, fish-hook — inside-out placement, no back-walling** (indigo carmine stains indistinct mucosa) → **three anterior 10-0 mucosal sutures** (abdominal lumen momentarily dilated first; small mucosal bite, $\frac{1}{3}$ – $\frac{1}{2}$ muscle; identical bites; exit through dot centre) → tie → **two 9-0 deep muscularis** between, not through mucosa → rotate 180° → **three posterior 10-0**; **heparinised Ringer's irrigation before last tie** → **four 9-0 deep muscularis** between mucosal sutures → 4–6 interrupted 9-0 nylon **adventitial layer** → **vasal sheath with 6–8 interrupted 8-0 nylon**, covering and de-tensioning

MULTILAYER VERSUS TWO-LAYER CLOSURE

Multilayer microdot	Two-layer alternative
10-0 mucosa → 9-0 deep muscularis → 9-0 adventitia → 8-0 sheath	10-0 mucosa → interrupted 8-0 nylon incorporating muscularis and adventitia between mucosal sutures
Deep muscularis sutures seal gaps without trauma from the 100-μm cutting needle	Fewer overlapping sutures; tension then taken by 5-0 Prolene sheath sutures

Reinforcing Vasal Suture Technique

- **Reinforcing vasal suture technique (ReVas)**: tension relief by reinforcing sutures; **markedly higher success and pregnancy rates**.
- Leave **perivasal sheath intact** during dissection → after anastomosis, sheath **3 cm proximal and distal reapproximated with 5-0 Prolene** into a **hairpin loop** → 5–10 further reinforcing sutures.

Anastomosis in the Convoluted Vas

- Technically harder; **fear of cutting back leads surgeons to anastomose in the straight portion onto an unhealthy testicular end** — poor blood supply, friable mucosa, gritty muscularis.

FIVE PRINCIPLES IN THE CONVOLUTED VAS

Principle	Detail
Perfect transverse cut	Round mucosal ring, lumen straight down; oblique lumen with thin flap unacceptable ; recut at 0.5-mm intervals (often 2–3 times) with 2.5–3 mm slotted clamp
Do not unravel	Unravelling disturbs blood supply at the anastomosis
Free sheath from epididymal tunica	Preserves supply; provides length
Small muscular/adventitial bites	Avoid perforating adjacent convolutions
Reinforce	Straight-portion sheath to convoluted sheath with interrupted 5-0 Prolene

Crossed Vasovasostomy

- Indications: **unilateral inguinal vasal obstruction with contralateral atrophic testis** → join healthy testis to contralateral unobstructed vas; **inguinal vas or ejaculatory duct obstruction/aplasia on one side with epididymal obstruction on the other**.
- Principle: **one anastomosis with high probability of success beats two operations with low probability** (e.g. unilateral vasoepididymostomy plus contralateral TURED).

Vas of atrophic testis transected at **straight-convoluted junction**; patency confirmed by saline or indigo carmine vasogram → contralateral vas dissected toward inguinal obstruction, clamped and transected **as high as possible** with right-angle clamp → testicular end crossed through **capacious opening in scrotal septum** → vasovasostomy as standard

- **Much easier than inguinal vasovasostomy**, which requires finding both ends in dense inguinal scar.

Intra-Abdominal Robotic Vasovasostomy

- For vas transected **above the inguinal ligament**, most often during **hernia repair** in children or adults; first reports 2014, robotic and laparoscopic.
- Insufflation → identify and mobilise vas → **modified Prentiss rerouting** for length; single-layer 8-0 or two-layer 9-0/10-0 anastomosis.
- Restrictions as for abdominal surgery: **no vigorous exercise or lifting >10 lb for 6 weeks**; semen analysis after 6 weeks.

Transposition of the Testis

- Critically short vas → **testicular transposition** for tension-free crossed anastomosis; **the spermatic cord is always longer than the vas**, so the testis crosses the septum comfortably.

Wound Closure

- **Drain rarely needed**; Penrose from dependent scrotum if extensive dissection, diffuse dartos oozing or **anticoagulation** — placed with care not to disturb anastomosis; patient removes it next morning.
- Dartos running 3-0 Vicryl incorporating septum; skin running horizontal mattress 5-0 Monocryl — midline raphe scar virtually invisible.

Postoperative Management

- Fluffs and scrotal support; **perioperative antibiotics only**; 2-week long-acting non-steroidal anti-inflammatory drug (NSAID) such as meloxicam, paracetamol, 3-day opioid supply.
- Scrotal support 4 weeks. **Ejaculation after 2 weeks; intercourse after 4 weeks**.
- Semen analysis at **6 weeks** and at **6 months** if no conception. **Poor motility at 6 weeks → 4-week prednisone course**, repeat analysis, repeatable once or twice.
- **Azoospermia at 6 months after vasovasostomy → redo vasovasostomy offered**.

POSTOPERATIVE RESTRICTIONS COMPARED

	Vasovasostomy	Vasoepididymostomy	Intra-abdominal VV	Varicocelectomy
Heavy lifting / strenuous	4 weeks (no >10 lb, squatting, running, cycling)	6 weeks	6 weeks (>10 lb)	All activity 4 weeks; 6 weeks if external oblique opened (>10–15 lb)
Ejaculation / intercourse	2 weeks / 4 weeks	2 weeks / 4 weeks	—	1 week

	Vasovasostomy	Vasoepididymostomy	Intra-abdominal VV	Varicocelectomy
First semen analysis	6 weeks	6 weeks	6 weeks	—

Postoperative Complications

- **Most common: haematoma** — rare, small, walnut-sized, perivascular; none needed drainage; resolve in 6–12 weeks. Wound infection did not occur.
- **Anastomotic sperm granuloma** — usually a harbinger of eventual obstruction.
- **Late stricture and obstruction are disappointingly common: progressive loss of motility followed by falling counts** signals stricture.
- Late obstruction reduced by **nylon instead of Prolene**, the microdot system, cutting back to healthy tissue, blood supply preservation and scrotal support until pregnancy.
- **Cryopreserve semen as soon as motile sperm appear** in the ejaculate.

Long-Term Follow-Up Evaluation After Vasovasostomy

- With sperm in the vasal fluid on at least one side, this technique returns sperm to the ejaculate in **virtually all** patients.
- Pregnancy is achieved in about half of couples, more when female factors are excluded; outcome depends on **interval since vasectomy and female partner age**.

KEY POINTS

- The vas deferens should not be stripped of its sheath.
- For a very large gap, gain length by freeing the convoluted vas from the epididymal tunica and by blunt dissection to the external ring; the convoluted vas should not be unravelled.
- Success relies on accurate mucosa-to-mucosa approximation, a tension-free and leakproof anastomosis, good blood supply, healthy mucosa and muscularis, and atraumatic technique.
- The ReVas technique creates a reinforcing loop of perivascular tissue, giving a tension-free anastomosis and higher success.

Concept — An anastomosis between two tubes without clotting factors is **only as good as its tissue, tension and mucosal sutures** — every step, from preserving the sheath to recutting gritty muscle to the ReVas loop, serves those three.

SURGERY OF THE EPIDIDYMIS

- Sperm acquire motility and fertilising capacity along a **single, tightly coiled tubule of about 200 µm diameter and 12–15 feet length**.
- After vasoepididymostomy even **short epididymal lengths adapt**; improvement may continue for **up to 2 years** — yet **preserving the greatest functional length gives the best sperm**.
- The wall is **thinnest at the caput and thickens distally** with more smooth muscle → **distal anastomoses are technically easier and more successful**.
- Corpus and cauda are one tubule → **injury anywhere obstructs at that level; magnification is essential** — loupes for macrodissection, microscope for anastomosis.

- Rich supply: testicular vessels superiorly, deferential inferiorly; **either — but not both — may be divided**. Epididymal branches lie **medial to and separate from** the main testicular artery → epididymal surgery need not compromise the testis.

Vasoepididymostomy

- Microsurgery converted a gross vas-epididymis reapproximation into a **precise vas lumen to single epididymal tubule** connection, markedly improving patency and pregnancy.
- **Highly specialised; only for experienced microsurgeons who perform it frequently.**

Indications

- At vasectomy reversal: per vasal fluid criteria (Table 70.2).
- Non-vasectomy OA: **complete spermatogenesis in at least one testis** (prior biopsy or **highly positive serum antisperm antibodies**) + **no sperm in vasal lumen** at exploration + **no vasal obstruction or EDO**. Preoperative evaluation as for vasovasostomy.

Microsurgical End-to-Side Vasoepididymostomy

- **Minimally traumatic, relatively bloodless, does not disturb epididymal blood supply**; clearly demarcated level (dilated above, collapsed below) makes site obvious; precise approximation of vas muscularis/adventitia to a tailored tunic opening.
- **Preferred when combined with inguinal vasovasostomy**: preserves the vasal supply from epididymal branches of the testicular artery, feeding the segment between the two anastomoses.
- Also preserves the **deferential contribution** when testicular artery integrity is doubtful — prior **orchidopexy, non-microscopic varicocelectomy, hernia repair**.

4 cm vertical median raphe incision, testis delivered → vas isolated with vas clamp + vessel loop at straight-convoluted junction → abdominal vas under tension; **anterior sheath and lumen cut with No. 11 blade in one swift stroke**; posterior sheath left intact → vasal fluid sampled — **no sperm confirms epididymal obstruction** → patency: 24-G sheath, 1–3 mL saline (indigo carmine rarely) → vas completely transected, prepared as for vasovasostomy → tunica vaginalis opened; epididymis inspected under microscope → site **above suspected obstruction, proximal to any granuloma, where dilated tubules show** → avascular tunic tented with jeweller's forceps; **3–4 mm round opening with iris scissors matching vas outer diameter** → sharp/blunt dissection to dilated loops → (level unclear: **puncture with 70-µm needle of 10-0 nylon, starting as distally as possible**, sample; when sperm found, seal with microbipolar and make new buttonhole just proximal) → abdominal vas tunnelled through separate opening in tunica vaginalis, fixed with **interrupted 5-0 Prolene** to perivasal adventitia and tunica albuginea — **critical tension relief**; vas must reach **with length to spare** → posterior epididymal tunic to posterior vas muscularis/adventitia with 2–3 interrupted 8-0 nylon

TABLE 70.3 COMPARISON OF THREE COMMON TECHNIQUES FOR VASOEPIDIDYMOSTOMY

Technique	Advantages	Disadvantages
Intussusception (LIVE)	Two longitudinal sutures in dilated tubule → four fixation points; virtually bloodless	Cannot assess tubular fluid for sperm before anastomosis set-up
End-to-side	Virtually bloodless; fluid examined before anastomosis	Difficult suture placement in a collapsed tubule
End-to-end	Fluid examined first; rapid identification of obstruction level; upward mobilisation bridges a large gap	Difficult haemostasis and outer closure; hard to identify correct tubule; inferior epididymal (vasal) supply sacrificed

- **What the table tests:** bloodlessness versus access — intussusception and end-to-side spare the epididymis; end-to-end buys length at the price of blood supply and precision.

Intussusception Vasoepididymostomy

- **Supplants end-to-side:** needles are placed while the tubule is still distended; the epididymal wall is **plastered against vasal mucosa**, and forward flow of epididymal fluid reinforces apposition → **more leakproof** than opening the tubule first.
- **Longitudinal intussusception vasoepididymostomy (LIVE)** is used for all cases: **four microdots** on the vas; **two parallel longitudinal sutures** in the distended tubule, **not pulled through** → tubule incised with 15° microknife exactly between them and sampled **without cutting sutures, leakage or collapse**.
- **Single-arm technique** almost as effective when double-armed sutures unavailable.

TRIANGULATION VERSUS LIVE INTUSSUSCEPTION

Original triangulation technique	Two-stitch LIVE
Three double-armed 10-0 sutures in a triangle	Two parallel longitudinal sutures
Opening torn in the centre with a 9-0 needle	Opening incised between sutures with 15° microknife
Original technique	Easier and more successful; now used for all cases

Technique When Vasal Length Is Severely Compromised

Vas prepared → tunica vaginalis opened, testis delivered → obstruction identified (**discrete yellow sperm granuloma**; indurated, dilated above; soft, collapsed below) or found by **distal-to-proximal puncture until sperm appear** → puncture sealed; epididymis **ligated just proximal with 6-0 nylon** → Penrose around epididymis at obstruction; under 2.5× loupes **dissected off testis 3–5 cm staying on tunica albuginea** → **inferior (and if necessary middle) epididymal branches ligated**; **superior branches at caput always preserved** → tunica vaginalis closed over testis with 5-0 Vicryl (**prevents drying and surface vessel thrombosis**); epididymis lies outside → dilated throughout: dissect to vasoepididymal junction (convoluted vas from below, epididymis from above) to keep maximal length → epididymis flipped up; **two-stitch end-to-side intussusception**

Varicocelectomy and Vasoepididymostomy

SIMULTANEOUS VARICOCELECTOMY: VASOVASOSTOMY VERSUS VASOEPIDIDYMOSTOMY

With vasovasostomy	With vasoepididymostomy
Acceptable , especially if intact deferential veins cross the anastomosis; microscope + artery preservation	Not recommended: post-varicocelectomy epididymal venous pressure is very high ; bleeding jeopardises a delicate, lower-success anastomosis
Or stage: reconstruct first, varicocelectomy ≥6 months later	Vasoepididymostomy first, wait 6 months; repair varicocele microsurgically if sperm quality poor and/or testosterone low

Long-Term Follow-Up Evaluation and Results

- In experienced hands sperm appear in the ejaculate in the **majority**; **intussusception patency exceeds that of classic end-to-side/end-to-end**.
- **Pregnancy is higher the more distal the anastomosis**, regardless of technique.
- Older methods: a substantial minority of initially patent anastomoses **shut down late**; with intussusception late shutdown appears lower, but long-term data are lacking.
- **Bank sperm intraoperatively and as soon as they appear in the ejaculate**, whatever the technique; banked sperm serve ICSI if counts stay poor or azoospermia persists.
- Persistent azoospermia without banked sperm → **redo vasoepididymostomy and/or sperm retrieval with IVF/ICSI**.

KEY POINTS

- Microsurgical vasoepididymostomy is a highly specialised procedure to be attempted only by experienced microsurgeons who perform it frequently.

Concept — *The epididymis is **one maturing tubule**: the more distal the anastomosis, the easier the suturing and the better the sperm — but it must sit **above the obstruction**, so every decision trades preserved length against reaching sperm.*

TRANSURETHRAL RESECTION OF THE EJACULATORY DUCTS

- **EDO is usually congenital** — the opposite end of the spectrum of excurrent duct anomalies that begins with **CBAVD and absence of most of the epididymis**; atresia confined to the terminal vas/ejaculatory duct is **correctable by transurethral resection (TUR)**.
- Acquired: prior **urethral instrumentation**, prior **genitourinary infection**, **extrinsic compression by prostatic or seminal vesicle cysts**. Higher ejaculatory duct pressures have been measured directly in EDO.

Diagnosis

- Suspect EDO in the **azoospermic or severely oligo- and/or asthenospermic man with at least one palpable vas, low volume, acidic pH and negative/equivocal/low fructose**; entertained if **FSH normal and biopsy shows normal spermatogenesis** (workup in Chapter 69).
- **Digital rectal examination (DRE)** may feel a midline cyst; **TRUS is key to both diagnosis and treatment** — midline cyst or dilated ducts and seminal vesicles.

TRUS-guided aspiration of cyst, dilated ducts or seminal vesicles (immediately before planned surgery; TRUS-biopsy bowel preparation and antibiotics) → motile sperm **cryopreserved** → **2–3 mL indigo carmine diluted with water-soluble contrast** instilled → radiograph shows resectable lesion → **TURED without prior vasography** (sperm in seminal vesicle = at least one epididymis patent and cyst communicates with an unobstructed vas); indigo marks the orifice and confirms opening

- **No sperm in aspirate** → **vasography**. **No sperm in either vas + EDO on vasography** → **abandon reconstruction**; microsurgical epididymal sperm aspiration or open TESE with cryopreservation. **Simultaneous vasoepididymostomy + TURED has never worked** in the authors' experience.
- EDO confirmed with **50% water-soluble contrast** and sperm in vasa → **24-G sheath left in place** for the assistant to inject dilute indigo carmine during resection.

TURED: Procedure

- **Cold-knife incision alone almost always reobstructs.**

24-Fr resectoscope loop; finger in rectum displaces posterior prostate anteriorly → ducts run from bladder neck to verumontanum, **exiting at the level of and lateral to the veru** → **resect the veru** to reveal dilated orifice or cyst cavity — preserving **bladder neck proximally, striated sphincter distally, rectal mucosa posteriorly** → **indigo carmine efflux confirms adequate resection**; avoid excessive coagulation → close hemivasotomies microscurgically if vasography done → **Foley overnight; 7 days oral antibiotics**

- Holmium laser ablation reported; **no comparative studies** of resection modalities.
- **Confirmatory vesiculography unnecessary** — high pressure can cause **secondary epididymal obstruction**.

Complications

COMPLICATIONS OF TURED

Complication	Features	Management
Reflux	Urine into ejaculatory ducts, vas, seminal vesicles after most resections ; urine impairs sperm quality	Document by voiding cystourethrography or semen creatinine
Epididymitis	Acute/chronic, chemical or bacterial; recurrent → epididymal obstruction ; incidence underestimated	Chronic low-dose antibacterial suppression (as for vesicoureteric reflux) until pregnancy ; chronic recurrent → vasectomy or epididymectomy
Retrograde ejaculation	Common even with bladder-neck care	Pseudoephedrine 60 mg 90 minutes before ejaculation , or Ornade Spansules (chlorpheniramine + phenylpropanolamine) twice daily for a week; midodrine, imipramine variable; else sperm from alkalinised urine for IUI or ICSI

Results

- Semen volume rises in most and **sperm appear in about half of previously azoospermic men**; pregnancy data are case reports and small series.
- Viable but poor-quality sperm → **IVF with ICSI**.
- Given potential serious complications, **TUR only in azoospermic or severely oligoasthenospermic men, and only after the couple declines IVF and is fully counselled**.

Seminal Vesiculoscopy

- Developed for **refractory haematospermia and ejaculatory duct stones**; **4–6 Fr scope through the prostatic utricle neck** → ducts entered and rinsed with saline; stones lasered and removed.
- Retrospective data: sperm return to the ejaculate in most within 12 months; **restenosis rates unknown; no comparison with TURED**.

TURED VERSUS SEMINAL VESICULOSCOPY

TURED	Seminal vesiculoscopy
24-Fr resection of veru and duct orifices	4–6 Fr endoscopy via utricle; rinse, laser stones
Destroys anti-reflux mechanism → reflux, epididymitis, retrograde ejaculation	Aims to avoid these risks
Established; results from small series	Restenosis unknown; no head-to-head data

Concept — *TURED trades obstruction for reflux: resection removes the valve that protected the excurrent ducts from urine, so its characteristic complications are the price of opening them — hence its restriction to couples who have declined IVF.*

SPERM RETRIEVAL TECHNIQUES

- Candidates: **CBAVD or bilateral partial vasal aplasia, failed or unreconstructable obstruction**, post-vasectomy men declining reconstruction, and **azoospermia (NOA)** — retrieval combined with IVF.
- **Intraoperative retrieval at reconstruction** (especially vasoepididymostomy, which has significant failure rates): used fresh if the partner is prepared, or cryopreserved for a later cycle.

- Sperm from chronically obstructed systems have **poor motility and fertilising capacity** → ICSI is **essential regardless of count and motility**.
- **SCI: fresh testicular sperm may give higher pregnancy rates than frozen-thawed.**

TABLE 70.4 SURGICAL TECHNIQUES FOR SPERM RETRIEVAL

Technique	Advantages	Disadvantages	Indication
MESA — microsurgical epididymal sperm aspiration	Lower complication rate; epididymal sperm more motile than testicular; large numbers for multiple cryopreserved vials	Microsurgical skill	OA; not NOA
PESA — percutaneous epididymal sperm aspiration	No microsurgery; local anaesthesia; motile epididymal sperm	Haematoma, pain, testicular vascular injury, epididymal obstruction ; variable success; fewer sperm than MESA	OA; not NOA
TESA — testicular sperm aspiration	No microsurgery; local anaesthesia	Immature/immotile sperm, small quantity ; haematoma, pain, vascular injury	OA; poor in NOA
TESE — testicular sperm extraction	Low complications if microsurgical	Microsurgical skill	Preferred for NOA

- **What the table tests:** epididymal methods give motile sperm in quantity but only where there is obstruction; testicular methods are the only route in NOA, and only microsurgical TESE does this well. **Open under vision beats percutaneous** at every site.

Microsurgical Epididymal Sperm Aspiration Techniques

Open Tubule Technique

- For **intraoperative retrieval at vasoepididymostomy** or as an **isolated procedure in CBAVD or unreconstructable obstruction**; median raphe for bilateral, otherwise transverse hemiscrotal.

Testis delivered, tunica vaginalis opened → epididymis inspected at **16–25×** → tunica incised over dilated tubule; bipolar haemostasis → tubule incised with **15° microknife** → fluid on slide + HTF + coverslip → no sperm: close tubule (10-0) and tunica (8-0 nylon), move **more proximally, up to the efferent ductules**, until motile sperm → **dry 5-μL micropipette** beside effluxing tubule; **capillary action only — never negative pressure** (disrupts mucosa); two pipettes speed recovery → flow highest just after incision; **quality improves after initial washout**; gentle testis/epididymal compression; **10–20 μL** recoverable → pipette on 3–5 cm silicone tubing (or butterfly tubing) → 20-G needle on Luer syringe flushes with 0.5–1.0 mL IVF medium → **discard each pipette** (residual fluid defeats capillary action); typically 4–12 → **cryopreserve in multiple aliquots**

- Haematocrit pipette less satisfactory; without micropipettes, **3-mL syringe with 24-G angiocath tip** filled with IVF medium, gentle aspiration.
- **Paradox: in obstructed systems motility is better proximally — the most motile sperm are often in the efferent ductules** (proximal — many sperm, occasional macrophages; distal — sperm fragments, macrophage-packed, grossly yellow tubules).
- **Fertilisation is highest with the longest epididymal length available** — the epididymis, even when debris-packed distally, secretes factors that diffuse proximally.
- With ICSI, **high ongoing pregnancy/delivery rates with fresh or cryopreserved epididymal sperm**; retrieval can be **elective** with vials for multiple cycles.

Percutaneous Epididymal Sperm Aspiration Technique

- **27-gauge tuberculin syringe with IVF medium**, multiple passes while aspirating.
- **Less reliable**; quantities sometimes too small to freeze; **pregnancy rates about half those of open techniques**; may **obstruct the epididymis** before a future vasoepididymostomy.
- Given the cost and effort of IVF, **retrieval under direct vision is preferred**.

KEY POINTS

- Puncture of a single tubule for sperm aspiration can be safely performed at the most proximal region of the caput epididymis.
- In obstructed systems sperm motility is better more proximally, with the most motile sperm found in the efferent ductules.

Testicular Sperm Extraction

- Indications: **no epididymal sperm despite spermatogenesis, or absent epididymis; NOA**.

THREE TECHNIQUES OF TESTICULAR SPERM RETRIEVAL

	Open microTESE	Percutaneous core biopsy	TESA
Instrument	Operating microscope	14-G biopsy gun	High-suction glass syringe, 23-G needle
Yield	Most sperm, least tissue	Intermediate	Often 10–20 passes needed
Invasiveness	Open	Percutaneous	Least invasive
Best for	NOA	OA with normal spermatogenesis	OA with normal spermatogenesis

Microsurgical Testicular Sperm Extraction

- The microscope finds an **avascular area of tunica albuginea** — protects blood supply and gives a blood-free specimen.
- **Schlegel's microdissection observation (1999)**: in NOA some tubules are larger, and **larger tubules are more likely to contain sperm**; biopsies show heterogeneity, and tubules with spermatogenesis are wider than Sertoli cell-only tubules.

SERTOLI CELL-ONLY VERSUS SPERMATOGENIC TUBULES

Sertoli cell-only	Active spermatogenesis
Thin, white, stringy	Larger, plumper, somewhat yellow
Low yield	Target for selective excision

General or regional anaesthesia → single median raphe or two transverse incisions in skin lines between scrotal vessels → testis delivered, tunica vaginalis opened → **10×: avascular plane on anterior tunica albuginea** → **generous midline transverse incision with 15° microknife** between vessels; crossing vessels coagulated first → testis opened **clam-shell** → tubules dissected with micro-needle holder or microbipolar forceps, **larger tubules selectively excised** with curved iris scissors → specimen in HTF, microdissected, **passed repeatedly through a 24-G angiocath sheath** to break tubules → examined at once by **andrology technician in theatre**; testis **searched systematically and mapped**; positive areas

recorded for any redo → negative → **contralateral testis** → microbipolar haemostasis → tunica albuginea closed 5-0 nylon or Prolene → tunica vaginalis continuous 4-0 Vicryl

- **Results:** sperm found in **about half of men explored**; partner pregnancy and live birth per ICSI cycle are modest, with a **high spontaneous abortion rate** — attributed to chromosomal abnormalities and DNA damage in NOA sperm.
- Sperm found and live births achieved even in **Sertoli cell-only syndrome, post-chemotherapy azoospermia and non-mosaic (47,XXY) Klinefelter syndrome**.
- **Flow cytometry with SYTO 17 red fluorescent nucleic acid stain** (binds only sperm) found sperm in most specimens deemed negative by light microscopy — **experimental; safety on human sperm unproven**.
- **Microfluidics:** proof-of-concept shows faster sperm search with minimal effect on morphology.

TABLE 70.5 TESE OUTCOMES BY DIAGNOSIS

Condition	Retrieval (%)
Cryptorchidism (post-orchidopexy)	74
AZFc deletion	70
Klinefelter syndrome	68
Post-chemotherapy	53
Maturation arrest	40
Sertoli cell-only	30
AZFa, AZFb deletions	0

Data from Chan et al., 2001; Hopps et al., 2003; Hung et al., 2007; Raman and Schlegel, 2003; Ramasamy et al., 2009; Ramasamy and Schlegel, 2007. Rows reordered by yield. AZF, azoospermia factor.

- **What the table tests:** genotype and history predict retrieval — **AZFa or AZFb deletion predicts zero retrieval** (do not operate), whereas AZFc deletion, Klinefelter and post-orchidopexy testes do well; **Sertoli cell-only is the worst histology but still not hopeless**.

KEY POINTS

- Open microTESE is the best technique in NOA.

Postmortem Sperm Retrieval

- First reported with **removal and mincing of the epididymis**, sperm cryopreserved, no pregnancies then; pregnancies now achieved with IVF/ICSI.
- **Vasal retrieval:** vas delivered → hemivasotomy with 15° microknife → testicular end cannulated with 22-G angiocatheter → irrigated with 0.2 mL HTF while convoluted vas and epididymis are massaged.
- **Ethics is the central issue:** current guidelines require that the patient gave permission for retrieval and use before death.

Concept — *In obstruction sperm production is intact, so any reservoir works and the epididymis is preferred for motility and quantity; in NOA production is focal, so retrieval becomes a search — which is why only microscope-guided selection of the larger tubules succeeds.*

VARICOCELECTOMY

- The most commonly performed operation for male infertility. Varicocele: about 15% of men, 35% with primary infertility, 75–81% with secondary infertility.

- Associated with a progressive, duration-dependent decline in testicular function.
- Repair halts further damage, improves spermatogenesis in many, improves live birth and IVF/ICSI outcomes and enhances Leydig cell function.
- The urologist's preventive role demands a technique that minimises complications and recurrence.

Scrotal Approach

- Scrotal drainage has deep and superficial networks; the deep network drains testis, epididymis and vas into the pampiniform plexus, intimately entwined with the testicular artery until it becomes a single testicular vein above the internal ring.
- The more distal the approach, the smaller the veins and the greater the risk to the artery; arterial injury defeats the purpose — atrophy, impaired spermatogenesis, chronic pain.
- Choice of approach is usually by surgeon preference and patient age.

Retroperitoneal Approach

- Incision at the level of the internal inguinal ring, external and internal obliques split, internal spermatic artery and vein exposed retroperitoneally near the ureter (Palomo approach).
- Advantages: proximal, near drainage into the left renal vein; only one or two large veins; artery not yet branched and often separate; fewest veins ligated; still common in children.
- High recurrence when the artery is preserved, especially in children and adolescents — failure from periarterial venae comitantes communicating with larger veins; less often parallel inguinal or retroperitoneal collaterals bypassing the ligation; cremasteric and scrotal collaterals cannot be reached.
- Artery (1.0–1.5 mm) hard to identify, especially in children; work is in a deep hole with vessels ligated in situ; lymphatics hard to preserve → hydrocele.
- In children/adolescents, intentional testicular artery ligation markedly reduces recurrence in children/adolescents (takes the periarterial veins) and reverses growth failure — but benefit to later spermatogenesis is uncertain and unlikely; in adults bilateral artery ligation has caused azoospermia and atrophy.

Laparoscopic Varicocelectomy

- In essence a retroperitoneal approach — same level as Palomo, similar pros and cons; recurrence from collaterals near or entering the renal vein.
- Magnification shows internal spermatic vessels and vas at the internal ring; artery preserved in most, lymphatics with experience; recurrence low in recent series but variable.
- Artery ligation with lymphatic sparing markedly reduces hydrocele in children.
- Complications more serious than open (bowel, vessel or visceral injury, air embolism, peritonitis) and general anaesthesia required; microsurgery needs local/regional and a 2–3 cm incision, no larger than the sum of port incisions; pain and recovery equal to subinguinal.
- Reasonable for bilateral varicoceles in experienced hands.

Microsurgical Inguinal and Subinguinal Operations: The Preferred Approaches

- Subinguinal microsurgical varicocelectomy is the most popular: cord delivered so artery, lymphatics and small periarterial veins are seen; access to external spermatic and gubernacular veins; access to testis for biopsy, epididymal inspection or hydrocele repair.
- Conventional (non-magnified) inguinal: 5 cm incision over canal, external oblique opened, cord delivered, internal spermatic veins ligated, vas and its vessels preserved, artery and lymphatics

sought, external spermatic veins ligated. Lowers recurrence versus retroperitoneal but not hydrocele or artery injury.

- Post-varicocelectomy hydrocele is due to lymphatic ligation (fluid analysis). Artery injury is probably under-reported; bilateral → azoospermia. Transplant series: dividing the cord and sparing only vas and vasal vessels → frequent atrophy and hydrocele.
- Microsurgery substantially reduces hydrocele (lymphatics seen) and allows preservation of the small testicular artery, avoiding atrophy and azoospermia.
- The claim that deferential and cremasteric arteries suffice is refuted: testicular artery diameter exceeds both combined; one-stage Fowler–Stephens orchidopexy (artery deliberately ligated) often atrophies; animal models — artery preservation improves, ligation worsens, testicular ultrastructure.

Anaesthesia

- Local 0.25% liposomal bupivacaine + oral sedation (oxycodone, midazolam); regional or light general if the testis is delivered.
- Skin and subcutaneous infiltration, then cord infiltrated proximally to avoid haematoma at the dissection site; blind cord block carries a small arterial risk → 30-gauge needle.

Inguinal and Subinguinal Approaches

- Subinguinal approach, just below the external ring: no fascia opened, less pain, faster recovery — but more veins, artery more often wrapped in a venous network and already divided into 2–3 branches, and pulsation dampened by the ring edge; topical papaverine helps.

TABLE 70.7 INDICATIONS FOR INGUINAL (EXTERNAL OBLIQUE OPENED) VERSUS SUBINGUINAL (FASCIA INTACT) VARICOCELECTOMY

Inguinal	Subinguinal
Prepubertal children	Prior inguinal surgery
Solitary testis	Obesity
Tight, low external ring	Lax, capacious external ring; high external ring
Short cord, high-lying testis	Long cord, low-lying testis
Less experienced microsurgeon	Very experienced microsurgeon

- What the table tests: open the fascia when the artery is precious or hard to find (child, solitary testis) or the ring is tight and low; stay subinguinal when the canal is scarred or hard to close, or the ring is already proximal.
- Always open the external oblique in prepubertal children without prior inguinal surgery — tiny artery, low blood pressure; exposure above branching gives clear pulsation.
- Prior inguinal surgery → subinguinal (cord adherent to external oblique undersurface). Prior failed subinguinal repair → consider opening the fascia to work proximal to scar.
- Subinguinal is significantly harder — for frequent operators; novices start inguinal. Simultaneous ipsilateral hernia repair → inguinal.

External ring located by invaginating scrotal skin and marked → incision sized to testis if delivery planned (atrophic 2–2.5 cm; larger 3 cm) in Langer lines: inguinal — from external ring laterally 2–3.5 cm; subinguinal — skin lines over the ring → Camper and Scarpa fasciae divided with cautery; superficial epigastric (superior) or superficial external pudendal (inferior) vessels retracted or, if central, divided and ligated → (inguinal) external oblique opened along fibres to ring; 3-0 absorbable at apex for closure → cord delivered with Babcock, ilioinguinal nerve and genital branch of genitofemoral excluded, encircled with large Penrose → (subinguinal) index finger along cord into

scrotum, hooked under ring to retract cephalad; small Richardson slid along back of finger and retracted caudad; cord delivered with Babcock → tongue depressor or knife handle inside Penrose as backboard; if Doppler signal poor, reduce cord tension

Dissection of the Cord

Microscope in → micro-Doppler maps number and location of arteries before dissection (handling causes vasospasm) → 6–10×: external spermatic fascia opened with Bovie on low (cut 5, coag 5) along cremasteric fibres (needle-point precise but can puncture veins; blunt tip cuts, coagulates, dissects) → internal spermatic fascia opened high and held with straight mosquitoes while testis pulled down → 10–25×, 1% papaverine irrigation, look for pulsation → micro-Doppler confirms → artery freed of tissue, tiny veins and lymphatics (microforceps, Jacobson mosquito, or toothless Adson/Gerald) → vessel loop → temporary occlusion under Doppler: signal disappears and returns — done before cutting any vessel ('measure twice, cut once') → if artery not obvious, start with largest veins, strip lymphatics, inspect undersides for adherent artery → all veins except vasal doubly ligated and divided → cord run over finger; small periarterial veins ligated or, if <1 mm, bipolar → end point

- Relying on visible pulsation is dangerous — superficial vessels transmit pulsation from deeper ones; the micro-Doppler is critical.
- The artery is commonly adherent to the undersurface of a large vein or encased in X- or H-shaped networks of small veins
- Vasal veins preserved; dilated vasal veins >2.5 mm are freed from the vasal artery and ligated. The vas carries two sets of veins — one intact set of deferential veins guarantees venous return.
- Cremasteric arteries (between two cremasteric veins) preserved in the great majority; power Doppler in NOA shows sperm-bearing tubules in the best-perfused regions → preserve maximal arterial supply.
- End point: only testicular arteries, cremasteric arteries, lymphatics and vas with its vessels remain; complete only when a run through reveals no further internal or external spermatic veins — each ligation makes the remaining veins dilate.

VEIN LIGATION BY CALIBRE

Vein diameter	Method
≥5 mm	Medium haemoclips
1–5 mm	Small auto-haemoclips
<2 mm	4-0 silk ligatures
<0.5 mm (<1 mm when periarterial)	Bipolar cautery
Vasal veins	Preserved ; ligated only if dilated >2.5 mm, one set kept

Delivery of the Testis

- Delivering the testis guarantees access to every route of venous drainage; cord delivery alone misses collaterals near the testis and scrotal/gubernacular collaterals, a radiographically proven cause of a minority of recurrences.
- Retrospective comparison of microsurgical inguinal versus subinguinal — no difference in operative time, semen improvement or pregnancy, although more veins and arteries were met subinguinally.
- Open microsurgical varicocelectomy has lower recurrence and fewer complications than laparoscopic or high retroperitoneal repair.

Upward traction on cord + upward pressure through invaginated scrotum → testis delivered → external spermatic veins doubly clipped and divided → gubernaculum inspected; veins leaving the tunica vaginalis cauterised or double-clipped → now all venous return lies within the Penrose-

encircled cord; run again, ligating residual veins → impulse test → external spermatic fascia may be closed over exposed arteries with 5-0 Vicryl (unnecessary)

- **Impulse test:** squeezing the dilated pampiniform plexus just above the testis transmits a distinct impulse to the largest veins on the testicular side of the ligatures — like a fluid thrill in ascites. No impulse → a significant internal, external or gubernacular vein has been missed → rerun cord, redeliver testis. It works because venous pressure stays high until deferential return develops.
- **Coexisting hydrocele** in a minority of varicocele testes; even a few millilitres alter testicular temperature regulation → repair if found: small — excise segment of sac and cauterise edges; larger — bottleneck or excision.
- **Transiently high venous pressure** after ligation hampers haemostasis → dependent scrotal drain for 24 hours after combined varicocelectomy and excisional hydrocelectomy.
- **Closure:** testis returned, drain beneath cord → external oblique (if opened) with the preplaced 3-0, excluding ilioinguinal and genital branch nerves → Scarpa/Camper 3-0 Vicryl → 5-0 absorbable subcuticular skin ± Steri-Strips → scrotal support with fluffs.
- **Aftercare:** most need no opioids; paracetamol and ibuprofen around the clock; ice in 20-minute periods; shower at 48 hours; desk work 2–3 days; ejaculation and intercourse at 1 week; light exercise 2 weeks; all activity 4 weeks; external oblique opened → no lifting >10–15 lb or vigorous exercise for 6 weeks (hernia risk).

Radiographic Occlusion Techniques

- **Embolisation** with mechanically occlusive solid or liquid agent under local anaesthesia and intravenous sedation; may take several hours.
- **Access:** right common femoral vein for left varicocele; internal jugular or basilic vein for right.
- **Failure** to cannulate small collaterals, external spermatic and scrotal collaterals → recurrence; technical failure in a proportion means many patients eventually need surgery.
- **Antegrade scrotal sclerotherapy** (Europe, via a scrotal vein): recurrence similar to balloon/coil; no long-term data; consequence of sclerosant escaping into renal vein and vena cava unknown; larger varicocele → more failure.

SOLID VERSUS LIQUID EMBOLIC AGENTS

Solid (coils, vascular plugs)	Liquid
Mechanical occlusion; thrombogenic fibres add thrombosis	Inflammatory endothelial necrosis and thrombosis
Risks: coil migration, venous perforation	Risks: catheter entrapment, non-target embolisation, glue migration, pampiniform phlebitis

TABLE 70.6 TECHNIQUES OF VARICOCELECTOMY

Technique	Artery preserved	Hydrocele (%)	Failure (%)	Serious morbidity
Retroperitoneal	No	7 (text: 7–33)	15–25 (text: ~15; 15–45 adolescents)	No
Conventional inguinal	No	3–30 (text: 3–15, mean 7)	5–15 (text: 9)	No
Laparoscopic	Yes	12	3–15 (text: recent series low, up to 17)	Yes
Radiographic	Yes	0	15–25 (text: balloon 4–11)	Yes
	Yes	0	0.5–1.0 (text: <1)	No

Technique	Artery preserved	Hydrocele (%)	Failure (%)	Serious morbidity
Microscopic inguinal / subinguinal				

Italic values in brackets are those given in the chapter body; table and body draw on different series and denominators.

- What the table tests: microsurgery is the only approach that is simultaneously artery-preserving, hydrocele-free, lowest-recurrence and low-morbidity; radiographic occlusion avoids hydrocele and artery injury but not recurrence; laparoscopy adds serious morbidity without lowering recurrence.

Complications of Varicocelectomy

COMPLICATIONS — MECHANISM AND PREVENTION

Complication	Mechanism	Prevention / notes
Hydrocele — most common after non-microscopic repair	Lymphatic obstruction (fluid protein analysis); many grow large enough to need excision; the insulating layer impairs countercurrent heat exchange , undoing the benefit	Magnification to preserve lymphatics almost eliminates it; treated as any hydrocele
Testicular artery injury	Artery 1.0–1.5 mm, supplies about two-thirds of testicular flow; often adherent to a large vein or in a venous network; spasm	Atrophy and/or impaired spermatogenesis; less in humans than animals (collaterals); children compensate better . Loupes/microscope + Doppler; radiographic occlusion eliminates this risk
Recurrence / persistence	Periarterial, parallel inguinal, mid-retroperitoneal, rarely trans-scrotal collaterals; commoner after paediatric repair	Retroperitoneal misses inguinal and scrotal; non-magnified inguinal misses scrotal and periarterial; microsurgery with testicular delivery lowest

- Diagnosing recurrence: veins may stay dilated and examination is unreliable → colour flow Doppler ultrasonography in the upright position showing flow reversal.
- Doppler is more sensitive than selective internal spermatic venography but has false positives (significance of low-grade reflux uncertain) and gives no anatomy; venography is usually done at embolisation — diagnostic and therapeutic.

Results

- Infertility with abnormal semen parameters → consider surgical varicocelectomy (AUA/ASRM guideline). Semen improves in most; pregnancy rates vary widely by series.
- Madgar trial (1995) — randomised controlled trial of surgery versus no surgery: pregnancy at 1 year 44% versus 10%.
- Large microsurgical series: high pregnancy rates by 1–2 years, higher when female factors excluded.
- NOA: no definitive evidence for varicocelectomy before assisted reproduction (AUA/ASRM); data are small uncontrolled series, and some NOA men show sperm on later analysis without intervention — interpret with caution.
- Large varicoceles improve more after repair but start worse → pregnancy rates similar regardless of size.
- Younger age at repair → greater improvement and recovery.
- Recurrence, artery ligation and postoperative hydrocele predict poor results.
- In infertile men with low testosterone, microsurgical varicocelectomy alone substantially raises serum testosterone.

Summary

- Varicocele is common and causes progressive, duration-dependent injury; larger varicoceles do more damage and gain more from repair; earlier repair recovers more; repair improves Leydig function, live birth and IVF/ICSI outcomes.
- The main complications — hydrocele, artery injury, persistence/recurrence — are reduced by microsurgery, an inguinal or subinguinal route and exposure of external spermatic and scrotal veins.

KEY POINTS

- Early repair of varicoceles prevents both future infertility and Leydig cell dysfunction.
- In select patients varicocelectomy may treat symptomatic, age-related androgen deficiency.
- The micro-Doppler is extremely useful in identifying arteries.
- Microsurgery through an inguinal or subinguinal approach decreases recurrence and hydrocele.
- As long as one set of deferential veins remains intact, venous return will be adequate.
- At completion only testicular arteries, cremasteric arteries, lymphatics and the vas with its vessels remain.
- Postoperative hydrocele is a direct result of failure to preserve lymphatic drainage.

Concept — Each complication maps to a structure mishandled — recurrence = a vein missed, hydrocele = a lymphatic tied, atrophy = an artery tied; the level of approach trades fewer veins against a less visible artery, and microsurgery with testicular delivery is the only approach that addresses all three.